SLIIT ACADEMY

# Higher Diploma
in
# Information Technology

Midterm Examination
Year 2, Semester 2 (2019)
January Intake

# Computer Security (IT1209)
# **Version A**

Duration: 1 Hour

CONFIDENTIAL

Instructions to Candidates:

- This is a closed book examination.
- This paper contains 20 MCQs (multiple-choice questions) on 6 pages without the cover page. Page 7 contains the supplementary document.
- Answer all questions on the paper itself.
- Read all questions before answering.
- The total marks obtainable for this examination is 20.

**Underline the correct answer from the following questions.** **(20 marks)**

1) Accurate and complete data enters the system for processing and remains accurate thereafter, is said to have:
   a) Confidentiality
   b) Availability
   c) Integrity
   d) Authentication

2) Encrypt "Attack at dawn " using Vigenère cipher. Use the keyword "magic". Use the Vigenère table given on page number 6. Underline the correct ciphertext from the following;
   a) MTZIEWAZLCIN
   b) MTZIEWAINZLC
   c) MTZWAINZLCIE
   d) MTZWALCINZIE

3) Assume you are an IT consultant for a business located in a coastal area that is susceptible to storms and occasional tsunami. Because of your company's location, there is an emphasis on continued business operation. Which of the following plans focus on ensuring that personnel, customers, and IT systems are minimally affected after a disaster?
   a) Risk management
   b) Fault tolerance
   c) Redundancy
   d) Business continuity

4) Which of the following types of security would be primarily concerned with Dumpster diving?
   a) Physical security
   b) Operational security
   c) Management and policy
   d) Authentication

CONFIDENTIAL

5) Upper management has suddenly become concerned about security. As the system administrator, you are asked to suggest changes that should be implemented. Which of the following access methods should you recommend if the method is to be allows a user to act in a certain predetermined manner based on the role the user holds in the organization?

a) MAC
b) RBAC
c) DAC
d) CHAP

6) A technician has travelled to a company's alternate site to assess current operations of the network infrastructure. Following the assessment, the technician has informed management that everything is operational, with all applications and data mirrored to re-establish service in a short time. Which of the following types of site does the company have?

a) Warm site
b) Cold site
c) Hot site
d) Standby site

7) How many bits exist in the secret key of DES (Data Encryption Standard) algorithm?

a) 56 bits
b) 32 bits
c) 48 bits
d) 16 bits

8) An administrator at a sister company calls to report a new threat that is making the rounds. According to him, the latest danger is an attack that attempts to intervene in a communications session by inserting a computer between the two systems that are communicating. Which of the following types of attacks does this constitute?

a) TCP/IP hijacking
b) Man-in-the-middle attack
c) Back door attack
d) Worm

CONFIDENTIAL

9) Which of the terms listed below applies to a collection of intermediary compromised systems that are used as a platform for a DDoS attack?

a) Quarantine network
b) Honeynet
c) Botnet
d) Honeypot

10) Which plan or policy helps an organization determine how to relocate to an emergency site?

a) Backup site plan
b) Risk management plan
c) Disaster-recovery plan
d) Business Continuity plan

11) Encrypt "This is a secret message" using rail fence cipher. Use the secret key as 3. Underline the correct ciphertext from the following;

a) TIIAERTESECRETCEMSAE
b) TIIAERTESGHSSSCEMSAE
c) TIETIHSSSCEMESSEIAREG
d) TIETSHSSSCEMSAEIAREG

12) Consider the following statements about Public Key Infrastructure (PKI).

A. PKI is a specific technology.
B. The main goal of PKI is to define an infrastructure that should work across multiple vendors, systems, and networks.
C. PKI is a single-key, symmetric system which use RSA for encryption.
D. PKI is a two key, asymmetric system which use RSA for encryption.

Which one is correct of the following?

a) Only A and B
b) Only B and D
c) Only A and D
d) Only B and C

CONFIDENTIAL

13) A security analyst is working on password-generation systems that require the implementation of an encrypted value from a mathematical process. Which of the following should the analyst use?

a) Hash function
b) Elliptic curve
c) Symmetric algorithm
d) Public key cryptography

14) An attempt to flood the network bandwidth by sending large streams of TFTP requests to a targeted system so that it becomes overwhelmed with false requests and in result cause a DoS situation is called:

a) TCP SYN flooding
b) TCP ACK flooding
c) UDP flooding
d) Session Hijacking

15) A technician wants to take a partial backup that stores only the information that has been changed since the last backup is known as;

a) Full backup
b) Incremental backup
c) Differential backup
d) Full Archival method

16) The attacker captured the information of the client and resubmits client's user credentials. What type of attack is this?

a) Spoofing
b) Social engineering
c) Trojan horse
d) Replay attack

CONFIDENTIAL

17) Harmful programs used to disrupt computer operation, gather sensitive information, or gain unauthorized access to computer systems are commonly referred to as:

a) Adware
b) Malware
c) Ransomware
d) Spyware

18) Decrypt "ZHCLTL" the message using Caesar cipher. Use the key value as 7 and the Caesar cipher table/ algorithms given on page number 6. Select the correct plaintext from the following;

a) URGENT
b) HELP ME
c) ESCAPE
d) SAVE ME

19) Your system administrator being informed to perform server backups. Which authentication method would be ideal for this situation?

a) MAC
b) DAC
c) RBAC
d) Security tokens

20) Bob wants to send a secret message to Alice, which key needs to be used to encrypt?

a) Bob's private key
b) Alice's private key
c) Alice's public key
d) Bob's public key

**End of the Question Paper**

CONFIDENTIAL

**Supplementary Documents**

Caesar Cipher

| a | b | c | d | e | f | g | h | i | j | k | l | m |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |

| n | o | p | q | r | s | t | u | v | w | x | y | z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

Encryption algorithm: $C = E(k, p) = (p + k) mod\ 26$

Decryption algorithm: $p = D(k, C) = (C - k) mod\ 26$

Vigenère Cipher Table

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| B | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
| C | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
| D | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
| E | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
| F | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
| G | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
| H | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
| I | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
| J | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
| K | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
| L | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
| M | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
| N | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
| O | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
| P | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
| Q | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| R | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
| S | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
| T | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
| U | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| V | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| W | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
| X | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
| Y | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
| Z | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |